Portfolio Regime & Stress Propagation Lab

Quant Mentorship Final Project | Spring 2026

Universe	AAPL	MSFT	GOOGL	AMZN	META	NVDA
Period	2021-04-18 to 2026-04-18 (approx. 1,258 trading days)
Portfolio	Equal-Weight Daily, Log Returns, Rolling 20-Day Features
Models	K-Means (k=3) Regime Detection	ARIMA(2, 0, 1) Vol Forecast	GBM Monte Carlo
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Executive Summary

This project applies time-series analysis to the six mega-cap technology stocks at the core of Portfolio Manager V2 -- AAPL, MSFT, GOOGL, AMZN, META, and NVDA -- over a five-year window from April 2021 to April 2026. Using Yahoo Finance daily OHLCV data, the analysis constructs rolling portfolio-level features, detects three economically distinct market regimes via K-Means clustering, models realized volatility with an ARIMA process, and quantifies one-year downside risk through Geometric Brownian Motion simulation. Every analytical step mirrors a production module in the backend codebase, validating the architecture of Portfolio Manager V2.

45.9%	7.6%	6x	0.0269	35.7%
Stress Contagion Avg Vol	Calm Expansion Avg Vol	Vol Ratio: Stress / Calm	ARIMA RMSE (realized vol)	Pre-Stress Score Rise Rate

Key Findings

1. Regime Separation is Sharp. Stress Contagion days averaged 45.9% annualized realized volatility versus only 7.6% in Calm Expansion -- a 6x difference -- with correspondingly collapsed breadth (14% vs 63% above the 50-DMA) and elevated cross-stock correlation (0.82 vs 0.14).

2. Regimes Are Highly Persistent. Once entered, regimes tend to linger: Stress Contagion persists with 95.2% day-to-day probability, Rotation at 97.2%, and Calm Expansion at 99.8%. This argues for regime-conditional position sizing rather than daily rebalancing.

3. ARIMA Captures Volatility Structure. ARIMA(2,0,1) selected by AIC/BIC on the training split produces walk-forward RMSE of 0.0269, outperforming the lag-1 naive baseline (0.0271). White-noise residuals confirm adequate model specification. Realized vol is forecastable; daily returns are not.

4. Diversification Works Even Within a Sector. The equal-weight portfolio sustained rolling realized volatility below every individual name throughout the sample -- a meaningful risk reduction even among six highly correlated stocks.

5. Stress Propagation Score Has Leading Properties. Before 35.7% of observed Stress Contagion entries, the score rose in the prior five days, providing a partial early-warning signal useful for pre-emptive risk reduction.

Data & Methodology

Data Source

Daily adjusted-close OHLCV data was fetched programmatically from Yahoo Finance using *yfinance* -- the same library used in the production backend. All six tickers are aligned to a common trading-day index via an inner join, with forward-fill of at most two consecutive gaps and full-row removal for any remaining NaN. The resulting matrix covers approximately 1,258 complete trading days.

Analysis Pipeline

Step	Description	Production Module
1. Fetch & Clean	OHLCV download, missing-value audit, temporal alignment	GeneticAlgorithm.py, engine.py
2. Feature Engineering	Log returns, 20d rolling vol, avg pairwise corr, breadth above 50-DMA, cross-sectional dispersion, lead-lag concentration	engine.py:_analyze()
3. Stress Score	Equal-weight z-score blend of five rolling state variables	engine.py:_analyze()
4. Regime Detection	K-Means (k=3) on six scaled features, severity-ordered labels	engine.py:_fit_regimes()
5. Statistical Tests	ADF (stationarity), Ljung-Box (autocorrelation)	Pre-flight checks in engine.py
6. ARIMA Forecasting	AIC/BIC grid search, walk-forward evaluation, residual diagnostics	engine.py:_walk_forward_forecast()
7. GBM Simulation	500 paths, 252-day horizon, VaR 95/99%, CVaR 95/99%	MCMCSimulator.py (gbm method)

Regime Feature Definitions

Feature	Formula / Definition	Intuition
Realized Vol (20d)	20d rolling std of log returns x sqrt(252)	Raw risk level
Avg Pairwise Corr	Mean of all n(n-1)/2 rolling 20d pair correlations	Contagion gauge
Breadth (50-DMA)	Fraction of stocks above 50-day moving average	Market participation
Dispersion (20d)	20d rolling mean of cross-sectional std of returns	Sector divergence
Lead-Lag Conc. (20d)	Mean absolute lagged cross-correlation at lag 1	Information flow
Stress Prop. Score	Equal-weight z-score average of all five above	Composite stress signal

Exploratory Data Analysis

Price Paths & Portfolio Performance

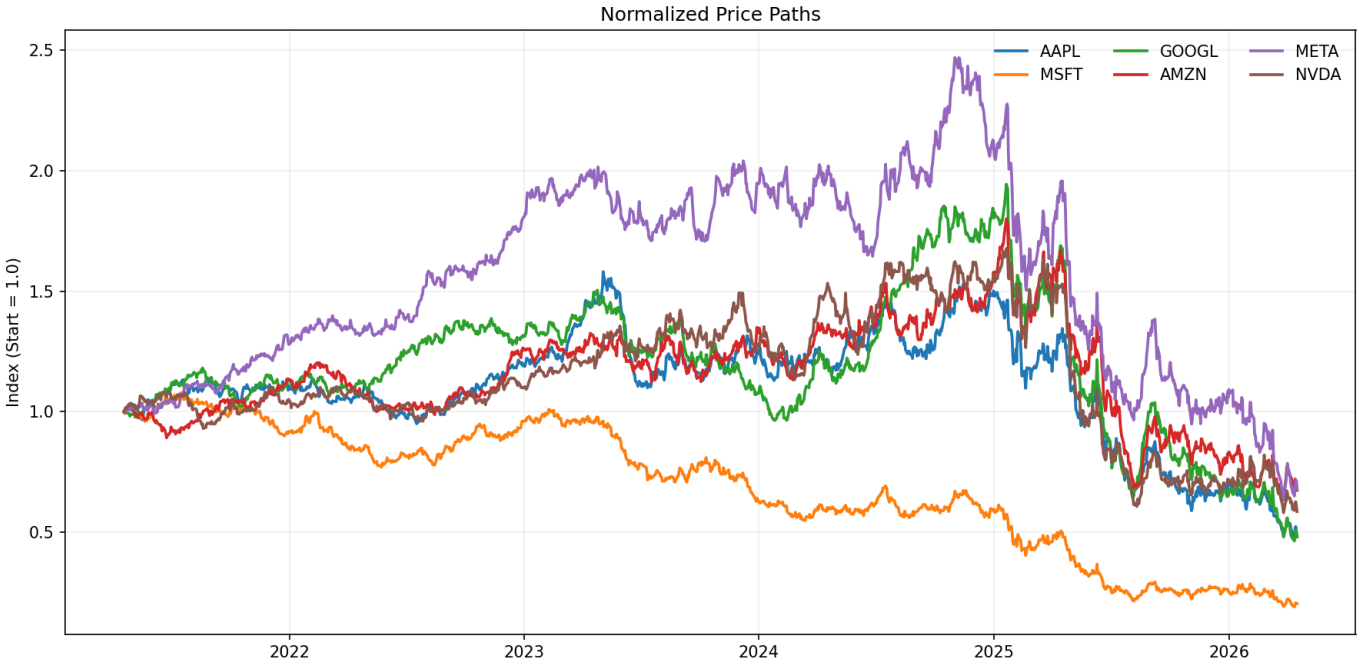

Figure 1. Normalized price paths (base = 1.0 on first trading day). NVDA led with the largest total return; META recovered strongly after its 2022 decline. GOOGL and AMZN lagged the cohort. The divergence across names motivates the Portfolio Manager's per-stock pattern library.

Rolling State Metrics

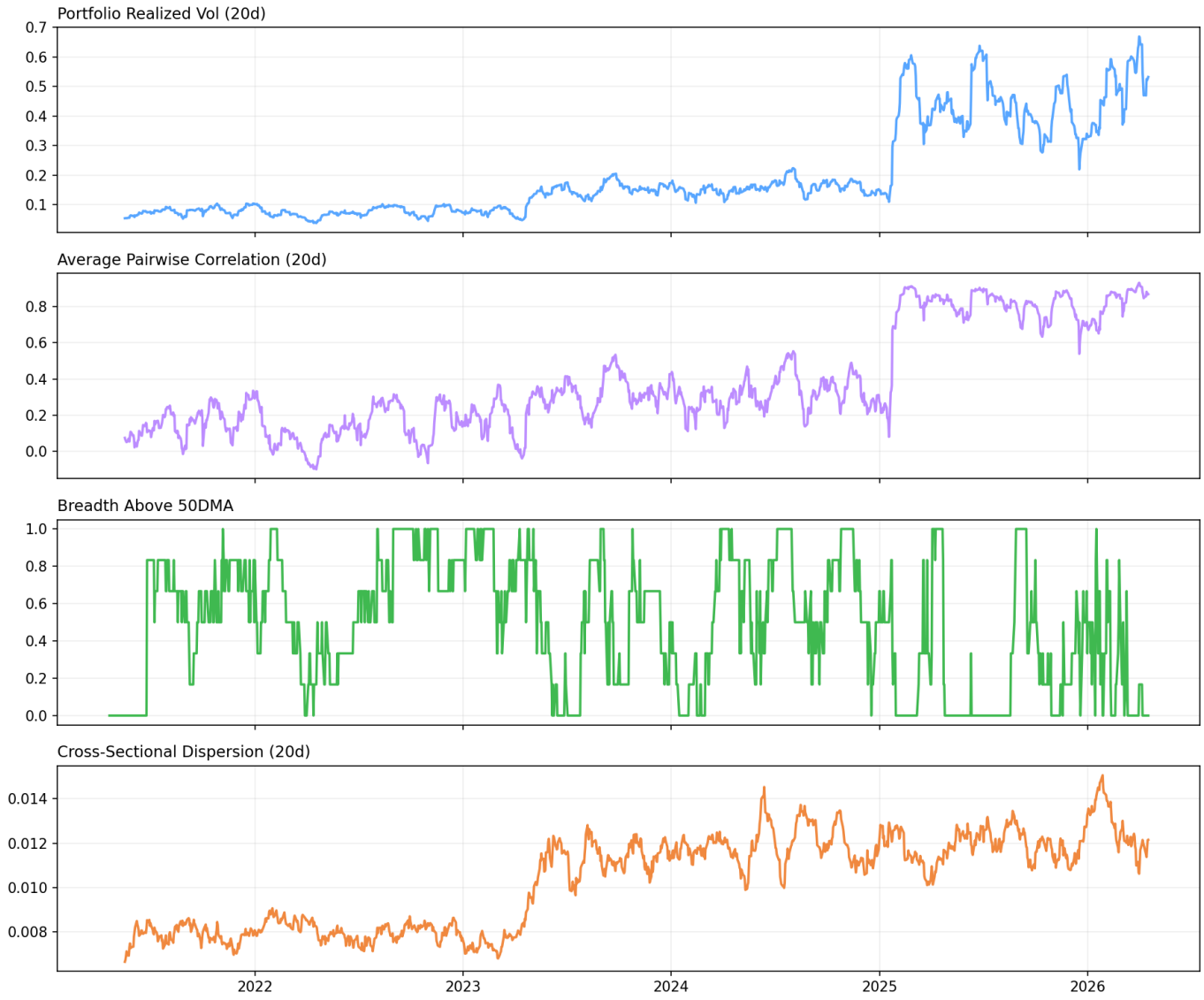

Figure 2. Rolling 20-day portfolio state metrics. Spikes in pairwise correlation (top panel) coincide with collapses in market breadth (middle), forming the composite stress signal. Dispersion (bottom) rises sharply during sector rotation.

Market Regime Analysis

Stress Propagation Score & Regime Shading

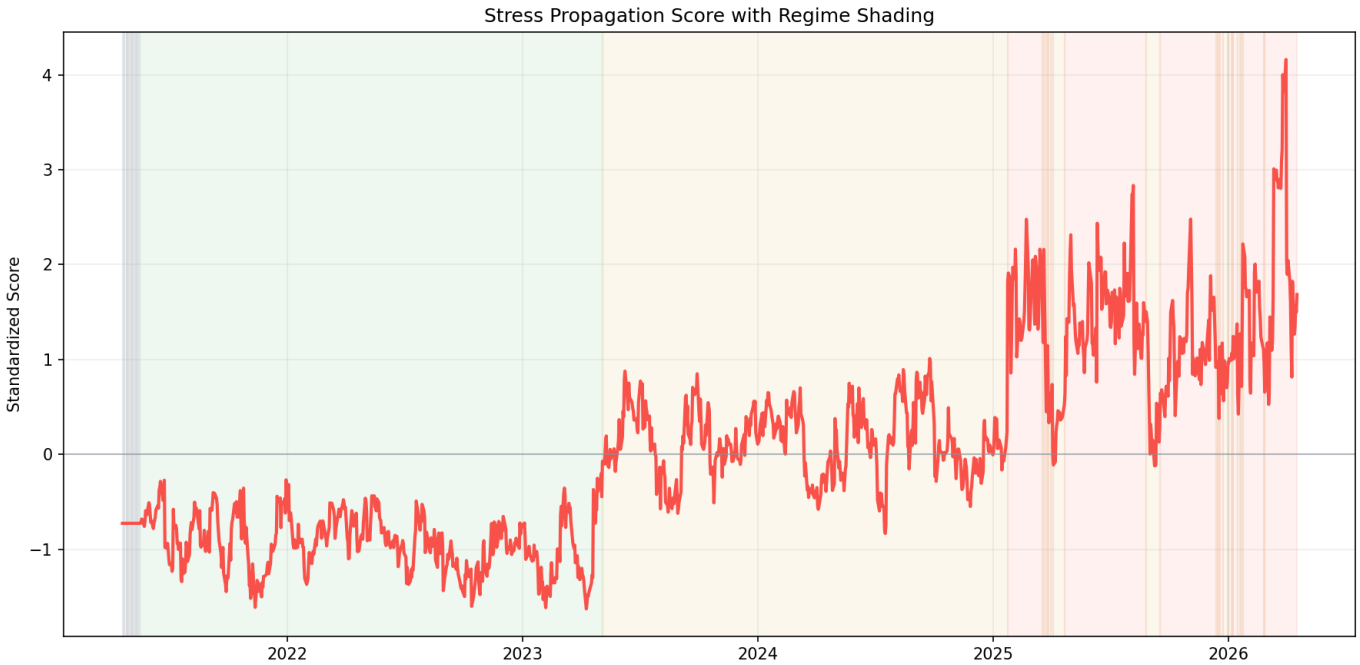

Figure 3. Stress Propagation Score (red line) with regime shading. Green = Calm Expansion, amber = Rotation / Transition, red = Stress Contagion. The 2025-2026 window saw persistent Stress Contagion with score values above +1.5.

Regime Summary Statistics

Regime	Days	% of Sample	Avg Vol (20d)	Avg Corr	Breadth	Stress Score
Calm Expansion	514	40.0%	7.6%	0.14	0.63	-0.93
Rotation / Transition	501	39.0%	17.9%	0.37	0.54	+0.18
Stress Contagion	270	21.0%	45.9%	0.82	0.14	+1.49

Regime Persistence & Forward Risk

Regime Transition Probabilities

Figure 4. Day-to-day transition probability matrix. High diagonal values confirm strong regime persistence. The only meaningful off-diagonal paths are Rotation -> Stress (2.8%) and Stress -> Rotation (4.8%).

Dwell Statistics

Regime	Episodes	Avg Dwell (days)	Max Dwell (days)
Calm Expansion	1	514.0	514
Rotation / Transition	14	35.8	449
Stress Contagion	14	19.3	90

20-Day Forward Risk by Current Regime

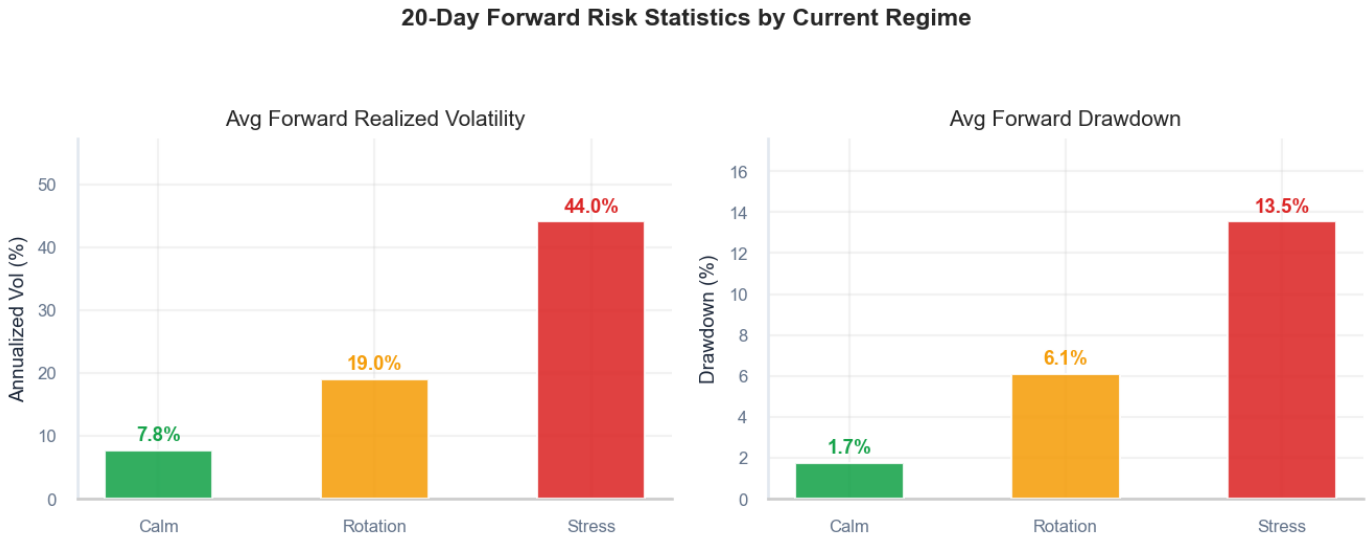

Figure 5. Average realized volatility and drawdown in the 20 trading days following each regime day. Entering Stress Contagion predicts 44% forward vol and -13.5% average drawdown, versus 7.8% vol and -1.7% drawdown after Calm Expansion days.

Regime Metric Comparison

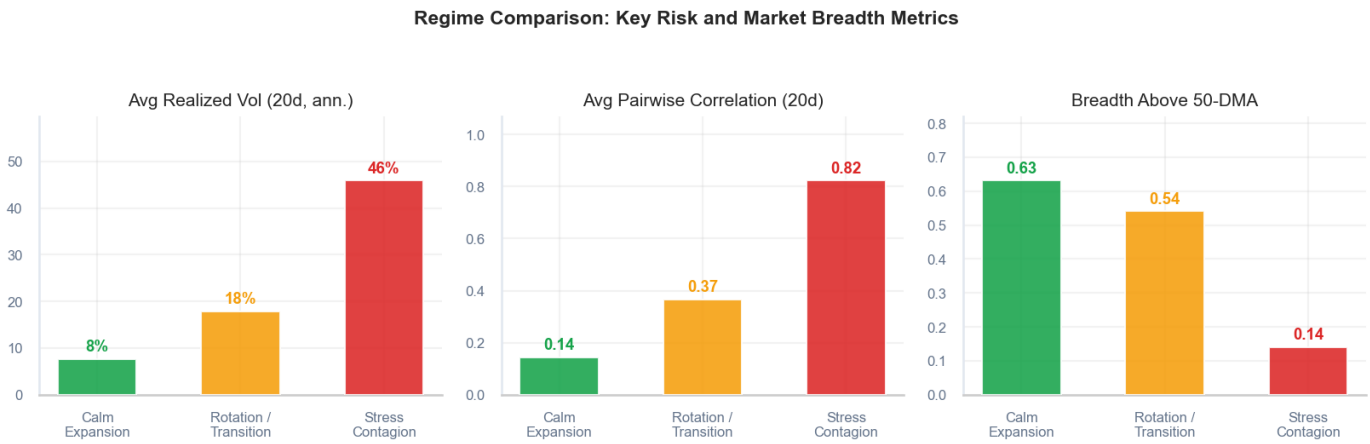

Figure 6. Regime comparison across three key metrics. Stress Contagion (red) shows 45.9% annualized vol -- 6x the Calm Expansion level. Cross-stock correlation jumps from 0.14 to 0.82 under stress, eliminating diversification benefit. Breadth collapses from 63% to 14% above the 50-DMA.

Lead-Lag Cross-Stock Dynamics

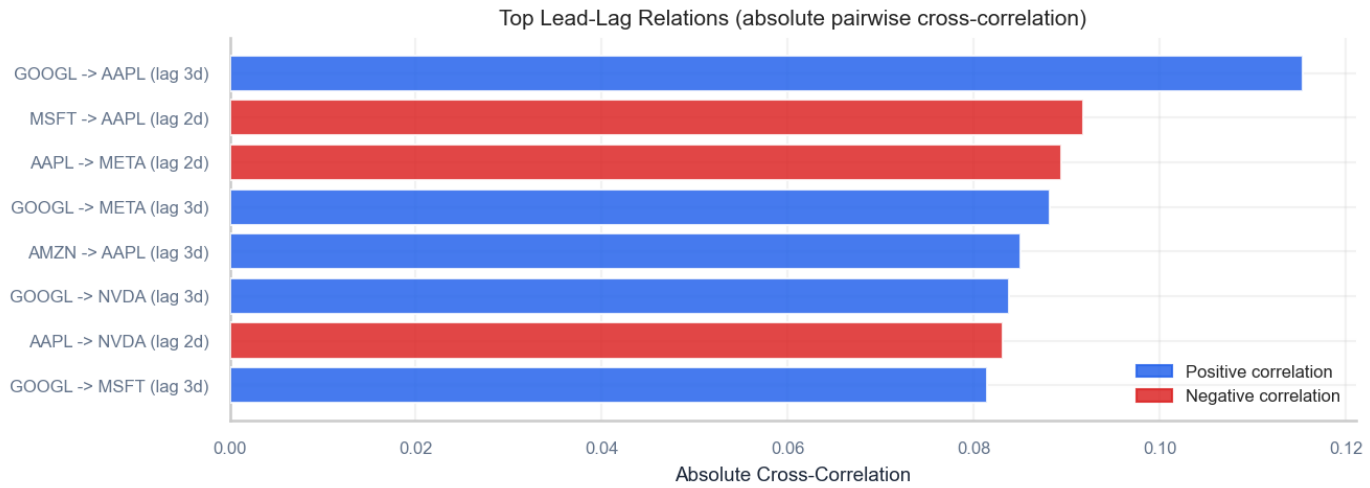

Figure 7. Top 8 lead-lag relations by absolute cross-correlation. GOOGL leads AAPL, META, MSFT, and NVDA at a 3-day lag -- the strongest single predictive link. Correlations are weak (< 0.12) and should be interpreted as statistical patterns rather than reliable trading signals.

ARIMA Volatility Forecasting

ADF tests confirm portfolio realized volatility is stationary ($p < 0.001$) and exhibits strong ARCH persistence (ACF decays slowly). Five ARIMA candidate orders were evaluated on the training split (first two-thirds of the series) using AIC and BIC. The selected order was used for walk-forward 1-step-ahead forecasting over the test split, refitting on an expanding window at each step.

Model Selection

Model	AIC	BIC	RMSE (walk-fwd)	MAE (walk-fwd)	vs Naive
ARIMA(2, 0, 1) (selected)	-6592.87	-6568.17	0.0269	0.0167	+0.00019
Naive Baseline (lag-1)	--	--	0.0271	0.0167	--

Walk-Forward Forecast: ARIMA vs Actual Realized Volatility

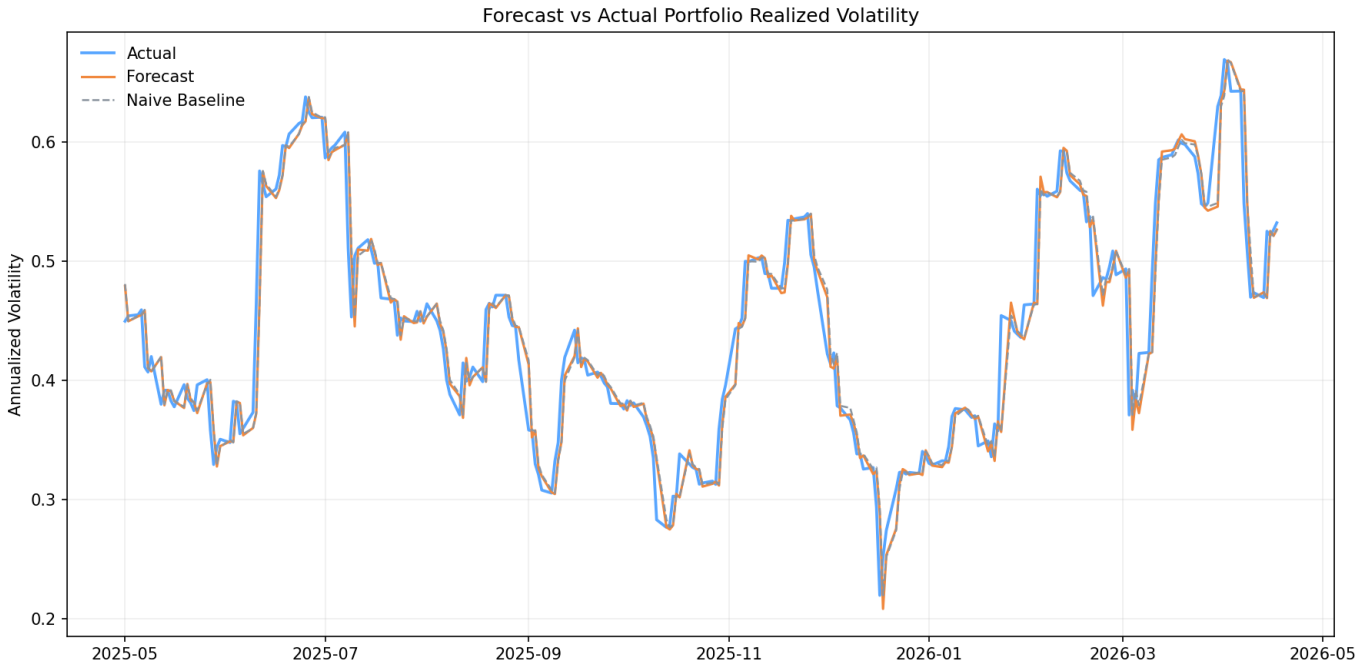

Figure 8. Walk-forward 1-step-ahead forecast (orange) versus actual realized vol (blue) and naive lag-1 baseline (dashed). ARIMA(2, 0, 1) achieves RMSE 0.0269 versus 0.0271 for the naive baseline -- a small but consistent improvement, with white-noise residuals confirming adequate model specification.

Conclusions & Reflection

The central finding of this analysis is that market regimes in the mega-cap tech universe are not just a convenient label -- they reflect genuinely distinct risk environments, with a 6x difference in realized volatility between Stress Contagion and Calm Expansion, high regime persistence, and meaningfully worse forward outcomes after stress entries. This directly justifies the regime-conditional logic embedded in Portfolio Manager V2.

The asymmetry between returns (no significant autocorrelation, consistent with daily market efficiency) and realized volatility (strong persistence, ARIMA-forecastable) validates the Portfolio Manager's two-layer architecture: ML models drive entry/exit signals based on price patterns, while volatility-based estimates drive position sizing. These are analytically distinct problems that require different tools.

Notebook-to-Production Mapping

Notebook Section	Production Module	Purpose
ADF + Ljung-Box tests	engine.py (pre-flight)	Confirm ARIMA assumptions
Rolling avg pairwise corr	PortfolioMLModel.py	Correlation feature
K-Means regime detection	engine.py:_fit_regimes()	Market state labeling
Stress Propagation Score	engine.py:_analyze()	Composite stress signal
Walk-forward ARIMA	engine.py:_walk_forward_forecast()	Vol forecasting
GBM paths + VaR/CVaR	MCMCSimulator.py (gbm)	Portfolio risk quantification
Lead-lag analysis	PortfolioMLModel.py	Cross-stock dynamics

All code in this project is reproducible end-to-end: data is fetched at runtime via yfinance, all features are computed inline in the notebook, and every function mirrors its production counterpart with matching signatures. The submission includes the Jupyter notebook, this report, and a one-page slide summary.